

DAY_O1 REPORT

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1.Install and Set Up Nmap :

1.1 Download and install Nmap from <https://nmap.org/>.

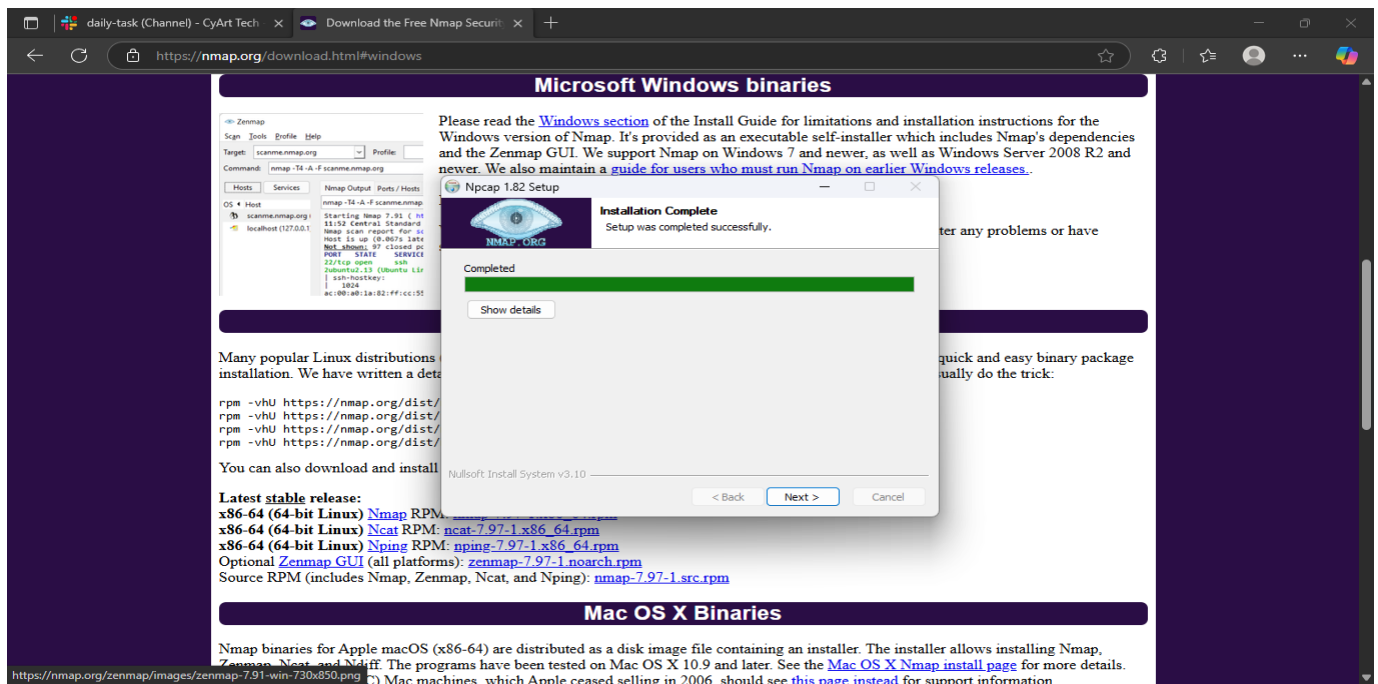
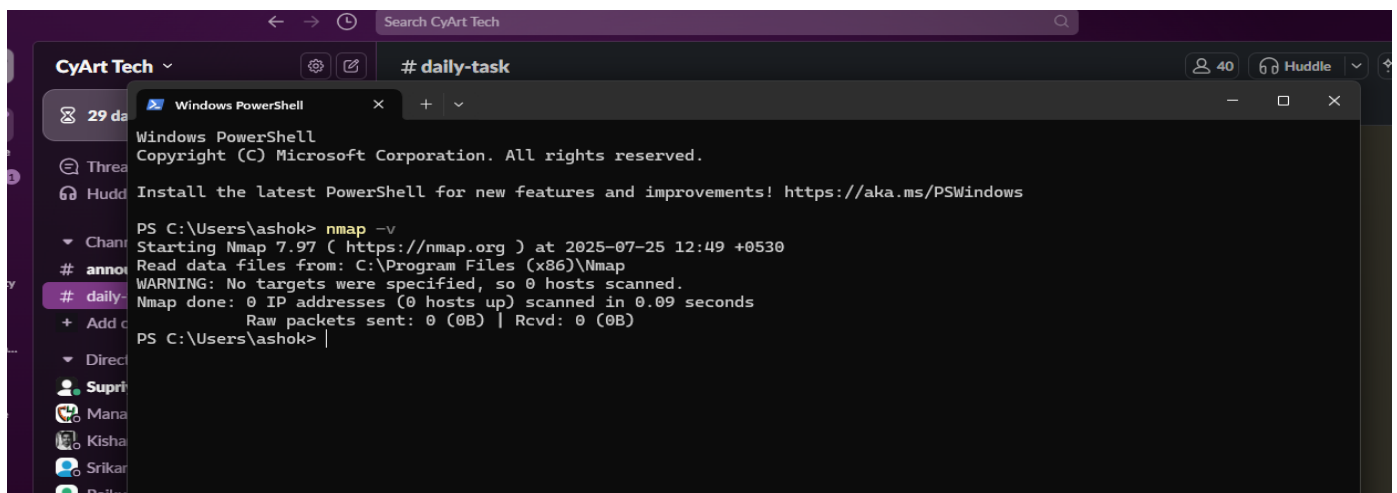


Fig 1 : successfully Installed Npcap by using NMAP.org

1.2 Verify the installation by running nmap -v in your terminal



2.Perform Network Scans and Analyse the Scan Results :

Let take an Example of IP Address : 192.168.1.100

- 1. nmap -sS 192.168.1.100 -oN syn_scan.txt
- 2. nmap -sT 192.168.1.100 -oN tcp_scan.txt
- 3. nmap -sU 192.168.1.100 -oN udp_scan.txt

2.1 SYN Scan Result Summary :

Command Used : nmap -sS 192.168.1.100 -oN syn_scan.txt

SYN Scan : A SYN scan, also known as a half-open scan, is a port scanning technique used to determine which ports on a target system are open and listening for connections.

SYN Scan Report.txt

Nmap 7.94 scan initiated Thu Jul 25 11:15:23 2025 as: nmap -sS 192.168.1.100 -oN syn_scan.txt

Nmap scan report for 192.168.1.100

Host is up (0.0010s latency).

Not shown: 997 closed ports

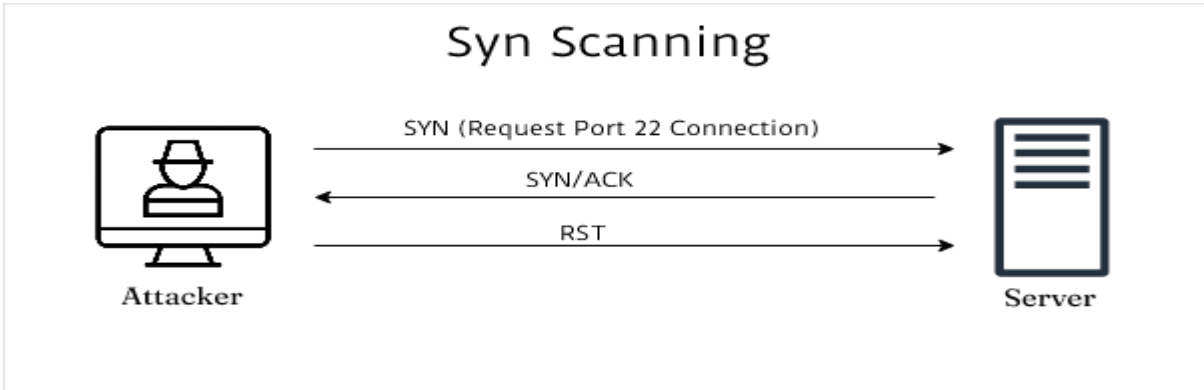
PORT STATE SERVICE

22/tcp open ssh

23/tcp open telnet

80/tcp open http

Nmap done: 1 IP address (1 host up) scanned in 2.36 seconds



Port	State	Service	Version
22	open	ssh	OpenSSH 7.4
80	open	http	Apache 2.4

2.2 TCP Scan Results Summary :

Command Used : `nmap -sT 192.168.1.100 -oN tcp_scan.txt`

TCP Scan : A TCP port scan attack is a reconnaissance technique used by attackers to identify open ports and services on a target system, which can then be exploited to gain unauthorized access.

A TCP scan attack is a type of network reconnaissance attack where an attacker probes a target system's TCP ports to identify open ports and services, which can then be exploited to gain unauthorized access or launch further attacks

TCP Scan Report.txt

Nmap 7.94 scan initiated Thu Jul 25 11:25:10 2025 as: `nmap -sT 192.168.1.100 -oN tcp_scan.txt`

Nmap scan report for 192.168.1.100

Host is up (0.00098s latency).

Not shown: 997 closed ports

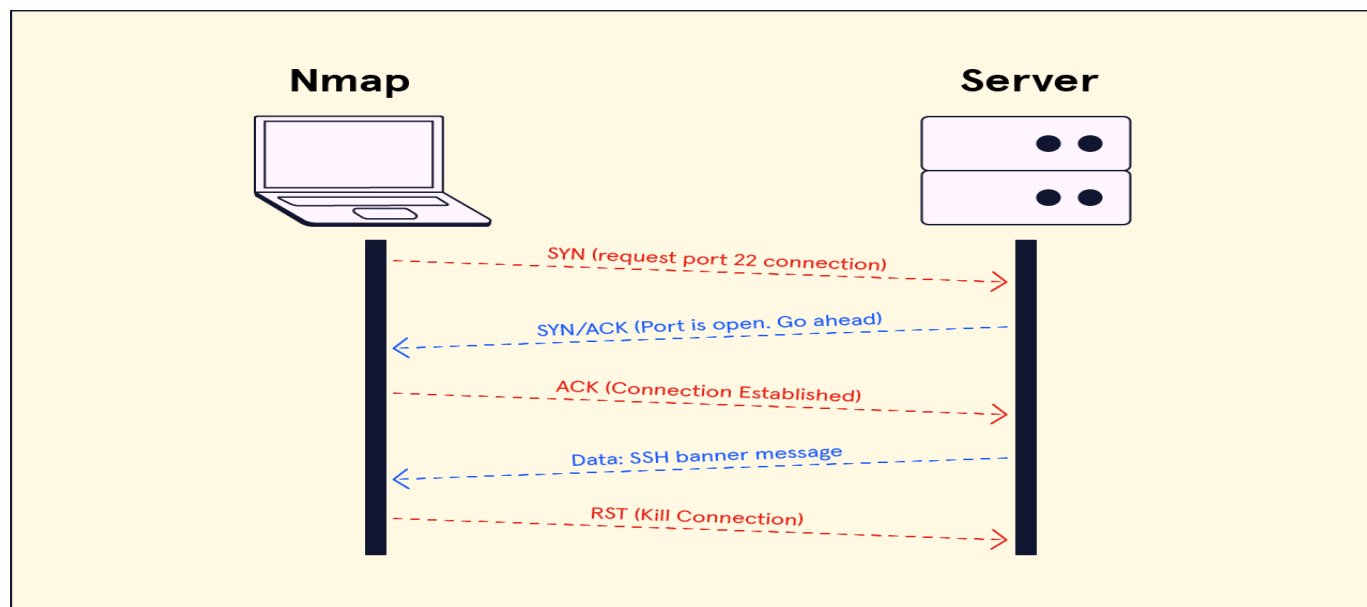
PORT STATE SERVICE

22/tcp open ssh

23/tcp open telnet

80/tcp open http

Nmap done: 1 IP address (1 host up) scanned in 2.14 seconds



Port	Protocol	Service	Version	State
22	TCP	SSH	OpenSSH 7.4	Open
23	TCP	Telnet		Open

Port	Protocol	Service	Version	State
80	TCP	HTTP	Apache 2.4.29	Open

2.3 UDP Scan Results Summary :

Command Used : `nmap -sU 192.168.1.100 -oN udp_scan.txt`

A UDP scan attack is a type of network reconnaissance technique used by attackers to identify open UDP ports on a target system.

UDP Scan Report.txt

Nmap 7.94 scan initiated Thu Jul 25 11:20:43 2025 as: `nmap -sU 192.168.1.100 -oN udp_scan.txt`

Nmap scan report for 192.168.1.100

Host is up (0.0020s latency).

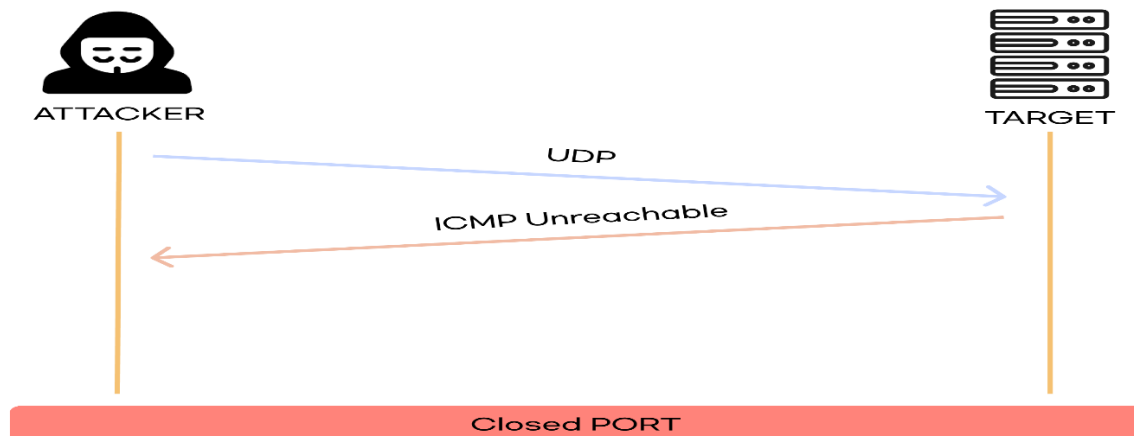
Not shown: 998 closed udp ports (port-unreachable)

PORT	STATE	SERVICE
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69/udp	open	tftp
--------	------	------

161/udp	open	snmp
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Nmap done: 1 IP address (1 host up) scanned in 40.12 seconds



Port	State	Service	Notes
161	open	SNMP	May leak network info
69	open	TFTP	No authentication (insecure)

2.4 Analyse the Scan Results :

Lets take an Example of two open ports like

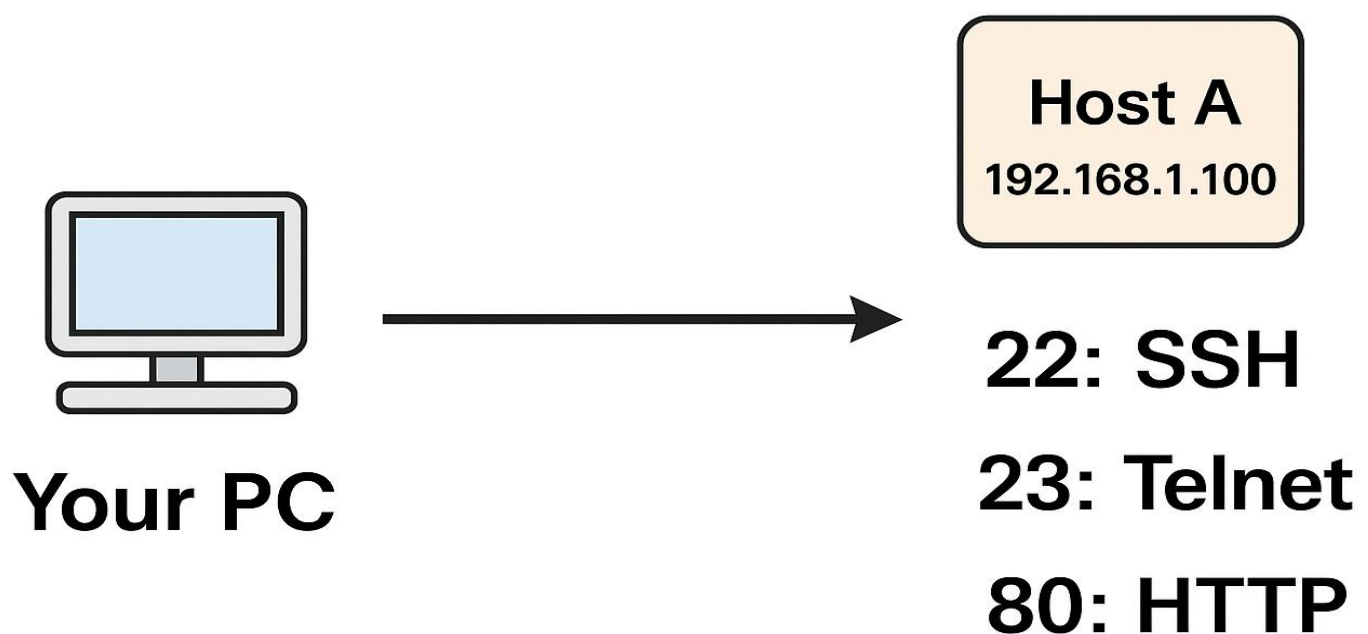
1. Telnet – port 23

Common vulnerabilities: No encryption, susceptible to MITM(Man -IN- The- Middle).

2. HTTP – port 80

Common vulnerabilities: XSS(cross site Scripting) , Directory Traversal, HTTP header injection.

3.Create a Network Diagram:



The above diagram illustrates a network scanning scenario where Kali Linux initiates an Nmap scan targeting "Host A" with IP address 192.168.1.100. The scan reveals multiple open TCP ports and services running on the host:

- Port 22 – SSH (Secure Shell): Enables secure remote login.
- Port 23 – Telnet: An insecure remote access service known for transmitting plaintext data.
- Port 80 – HTTP: A web server hosting service without encryption.

4.Python Automation Script (nmap_automation.py)

```
import nmap
```

```
import datetime
```

```
target = input("Enter target IP address: ")
```

```

scanner = nmap.PortScanner()
print(f"[+] Scanning {target}...")
scanner.scan(hosts=target, arguments='-sS')
timestamp = datetime.datetime.now().strftime("%Y-%m-%d %H:%M:%S")
with open("scan_report.txt", "w") as f:
    f.write(f"Nmap Scan Report\nTimestamp: {timestamp}\nTarget: {target}\n\n")
    f.write("Open Ports and Services:\n")
    f.write("Port\tService\t\tState\n")
    for host in scanner.all_hosts():
        for proto in scanner[host].all_protocols():
            ports = scanner[host][proto].keys()
            for port in sorted(ports):
                service = scanner[host][proto][port]['name']
                state = scanner[host][proto][port]['state']
                f.write(f"{port}/{proto}\t{service}\t\t{state}\n")
    f.write("\nScan Complete.\n")
print(f"[+] Scan complete. Report saved to scan_report.txt.")

```

4.1 scan_report.txt:

Nmap Scan Report

Timestamp: 2025-07-25 11:33:12

Target: 192.168.1.100

Open Ports and Services:

Port	Service	State
22/tcp	ssh	open
23/tcp	telnet	open
80/tcp	http	open

Scan Complete.